

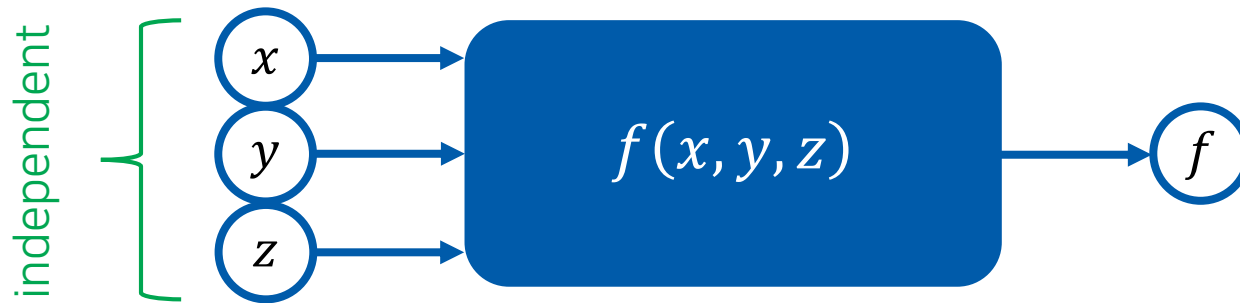
화학수학

네 번째 수업

다변수 미분

Function of Several Independent Variables

(다변수함수)



Example: $f(x, y, z) = x^2 + xy + 3z$

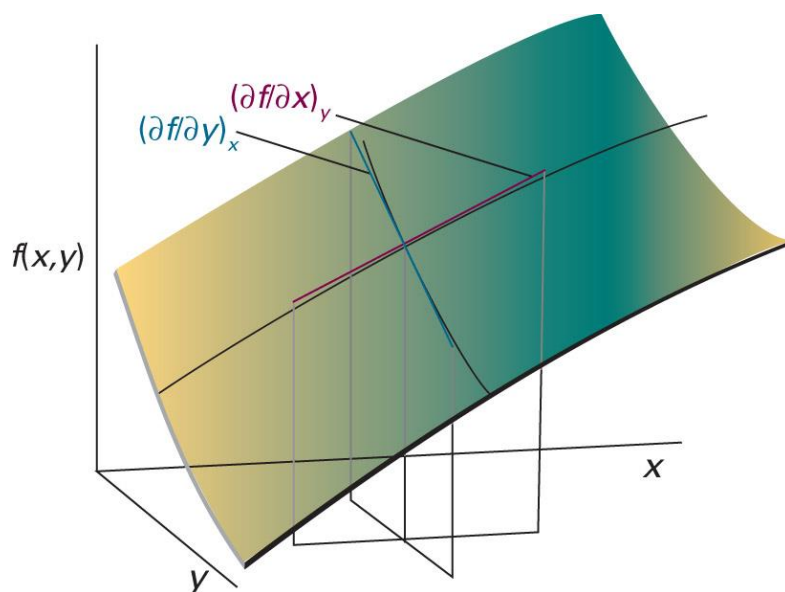
$$f(1, 1, 1) = 1 + 1 + 3 = 5$$

$$f(-1, 1, 0) = 1 - 1 + 0 = 0$$

⋮

Partial Derivatives (편미분)

“the slope of the function with respect to one of the variables,
all the other variables being held constant”



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Fundamental Equation of Differential Calculus

(미분학의 기본 방정식)

- n independent variables: $x_1, x_2, \dots, x_n$
- dependent variable: y

The differential of y is

$$dy = \sum_{i=1}^n \left(\frac{\partial y}{\partial x_i} \right)_{x'} dx_i$$

where x' means keeping all of the variables except for x_i fixed

Variable-Change Identity

(변수-변화 항등식)

$$\left(\frac{\partial U}{\partial T}\right)_{V,n} = \left(\frac{\partial U}{\partial T}\right)_{P,n} + \left(\frac{\partial U}{\partial P}\right)_{T,n} \left(\frac{\partial P}{\partial T}\right)_{V,n}$$

Reciprocal Identity

(역 항등식)

$$\left(\frac{\partial y}{\partial x}\right)_{z,u} = \frac{1}{(\partial x / \partial y)_{z,u}}$$

Euler Reciprocity Relation

(오일러 역관계식)

For $z = z(x, y)$,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Cycle Rule

(순환 법칙)

$$\left(\frac{\partial y}{\partial x}\right)_z \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = -1$$

Chain Rule

(연쇄 법칙)

For $z = z(u, x, y)$ and $x = x(u, v, y)$,

$$\left(\frac{\partial z}{\partial y}\right)_{u,v} = \left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v}$$

Note:

$$\left(\frac{\partial z}{\partial x}\right)_{u,v} \left(\frac{\partial x}{\partial y}\right)_{u,v} \left(\frac{\partial y}{\partial z}\right)_{u,v} = 1$$

Check the subscripts!

오늘의 수업

- Exact and inexact differentials
 - Test of exactness
 - Integrating factors
- Optimization
 - Constrained optimization
 - Lagrange's method
- [Review] Coordinate systems and vectors

Pfaffian Form: General Differential Form

(파피안형)

$$du = M(x, y)dx + N(x, y)dy$$

- Exact differential (완전 미분):

$$M(x, y) = \left(\frac{\partial u}{\partial x} \right)_y, \quad N(x, y) = \left(\frac{\partial u}{\partial y} \right)_x$$

- Inexact differential (불완전 미분): otherwise

Test of Exactness

$$du = M(x, y)dx + N(x, y)dy$$

Using the Euler reciprocity relation,

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

if the differential is exact,

$$\left(\frac{\partial N}{\partial x} \right)_y = \left(\frac{\partial M}{\partial y} \right)_x$$

Example 8.11

Show that the following differential is exact:

$$dz = \underbrace{\left(2xy + \frac{9x^2}{y}\right)}_{M(x,y)} dx + \underbrace{\left(x^2 - \frac{3x^3}{y^2}\right)}_{N(x,y)} dy$$

Using the previous criterion,

$$\left(\frac{\partial M}{\partial y}\right)_x = \left[\frac{\partial}{\partial y}\left(2xy + \frac{9x^2}{y}\right)\right]_x = 2x - \frac{9x^2}{y^2}$$

$$\left(\frac{\partial N}{\partial x}\right)_y = \left[\frac{\partial}{\partial x}\left(x^2 - \frac{3x^3}{y^2}\right)\right]_y = 2x - \frac{9x^2}{y^2}$$

Since the two terms are equal, the differential is exact.

Test of Exactness: Three Variables

$$du = M(x, y, z)dx + N(x, y, z)dy + P(x, y, z)dz$$

If the differential is exact,

$$\left\{ \begin{array}{l} \left(\frac{\partial M}{\partial y} \right)_{x,z} = \left(\frac{\partial N}{\partial x} \right)_{y,z} \\ \left(\frac{\partial N}{\partial z} \right)_{x,y} = \left(\frac{\partial P}{\partial y} \right)_{x,z} \\ \left(\frac{\partial M}{\partial z} \right)_{x,y} = \left(\frac{\partial P}{\partial x} \right)_{y,z} \end{array} \right.$$

Inexact Differentials in Thermodynamics

- δq = the amount of heat transferred to the system
- δw = the amount of work done on the system

For a reversible process, $\delta w_{\text{rev}} = -PdV$

Exactness test: for $\delta w_{\text{rev}} = MdT + NdV$,

$$\left(\frac{\partial M}{\partial V}\right)_T = 0, \quad \left(\frac{\partial N}{\partial T}\right)_V = -\frac{nR}{V} \neq 0$$

The differential is inexact

First Law of Thermodynamics

$$dU = \delta q + \delta w$$

dU is an exact differential
even though δq and δw are not

Integrating Factor

(적분인자)

inexact differential → exact differential
by multiplying a function (integrating factor)

No general method to find an integrating factor

Example 8.13

Show that the differential

$$du = Mdx + Ndy = (2ax^2 + bxy)dx + (bx^2 + 2cxy)dy$$

is inexact, but that $1/x$ is an integrating factor, so that du/x is exact.

First, test the exactness:

$$\left(\frac{\partial M}{\partial y}\right)_x = \left[\frac{\partial}{\partial y}(2ax^2 + bxy)\right]_x = bx$$

$$\left(\frac{\partial N}{\partial x}\right)_y = \left[\frac{\partial}{\partial x}(bx^2 + 2cxy)\right]_y = 2bx + 2cx$$

so the differential is inexact.

Example 8.13

Show that the differential

$$du = Mdx + Ndy = (2ax^2 + bxy)dx + (bx^2 + 2cxy)dy$$

is inexact, but that $1/x$ is an integrating factor, so that du/x is exact.

Now consider du/x :

$$du/x = (2ax + by)dx + (bx + 2cy)dy = Pdx + Qdy$$

so

$$\left(\frac{\partial P}{\partial y}\right)_x = \left[\frac{\partial}{\partial y}(2ax + by)\right]_x = b$$

$$\left(\frac{\partial Q}{\partial x}\right)_y = \left[\frac{\partial}{\partial x}(bx + 2cy)\right]_y = b$$



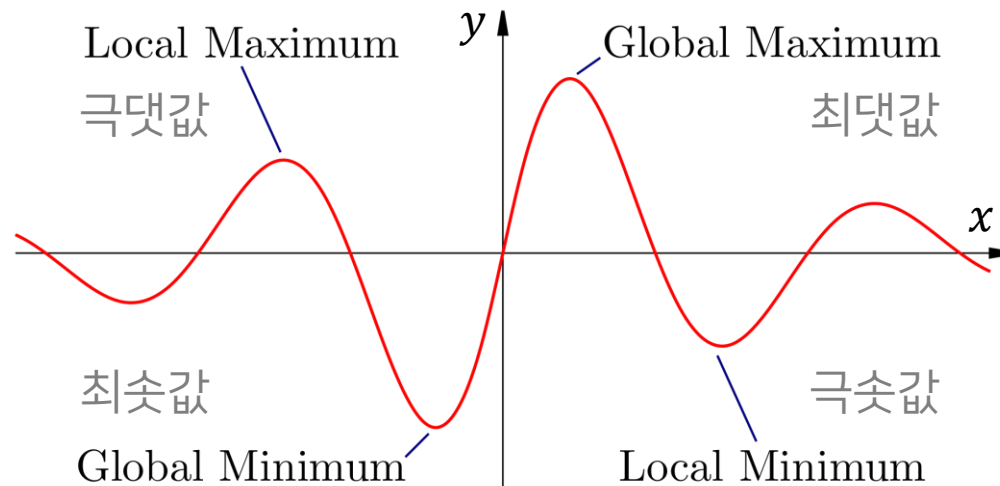
Hence, the differential is exact.

Second Law of Thermodynamics

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

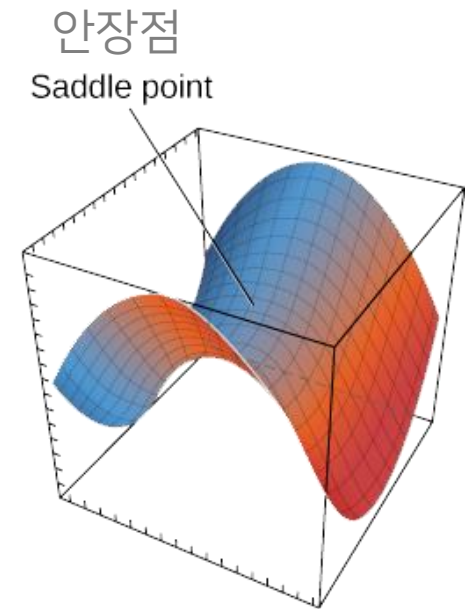
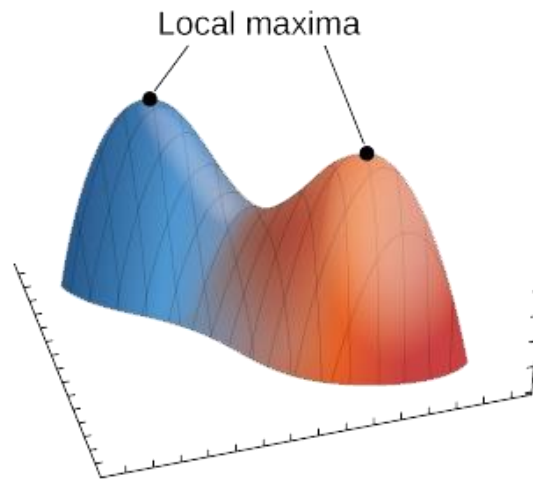
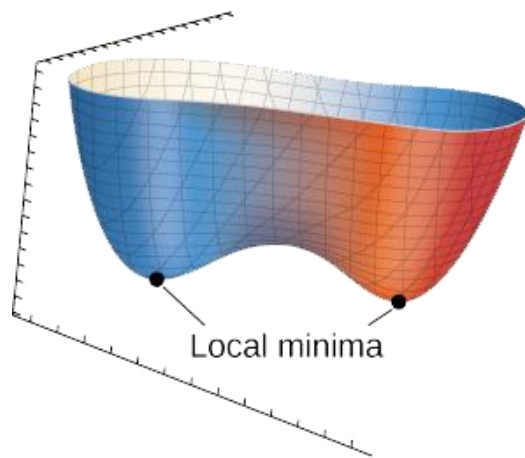
dS is an exact differential
even though δq_{rev} is not
(integrating factor: $1/T$)

Extrema in One Dimension



- At an extremum, $dy/dx = 0$
- At a minimum, $d^2y/dx^2 > 0$
- At a maximum, $d^2y/dx^2 < 0$

Extrema in Higher Dimensions



Finding Extrema in Two Dimensions

For $f = f(x, y)$,

1. Solve the simultaneous equations

$$\left(\frac{\partial f}{\partial x}\right)_y = 0, \quad \left(\frac{\partial f}{\partial y}\right)_x = 0$$

2. Check the value of the function at all points satisfying these equations, and at the boundaries, cusps, discontinuities (if any)

Example 8.14

Find the extremum value of the function $f = e^{-x^2-y^2}$.

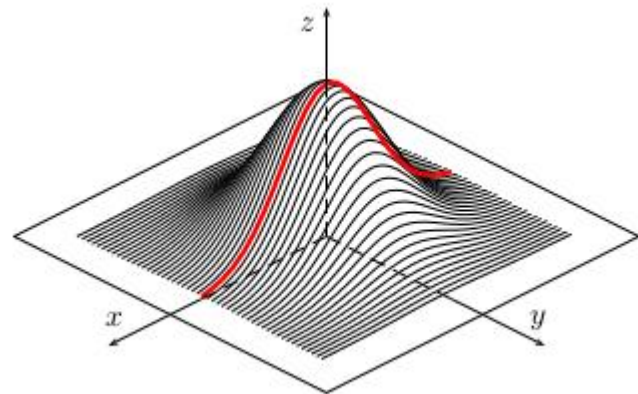
At a maximum or minimum,

$$\left(\frac{\partial f}{\partial x}\right)_y = -2xe^{-x^2-y^2} = 0$$

$$\left(\frac{\partial f}{\partial y}\right)_x = -2ye^{-x^2-y^2} = 0$$

The only solution is $x = 0$, $y = 0$.

Hence, the extremum value is $f(0, 0) = 1$.



Test for Identity of Extremum

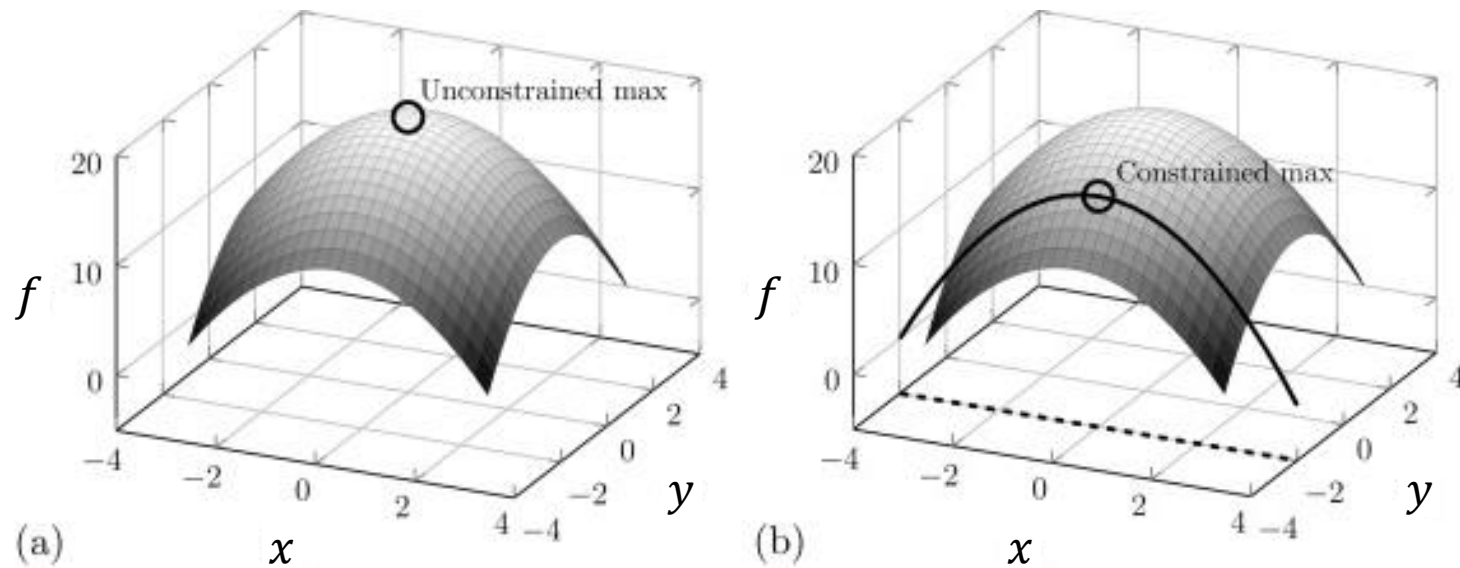
For $f = f(x, y)$, calculate the following quantity:

$$D = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2$$

1. local minimum: $D > 0$ and $\partial^2 f / \partial x^2 > 0$
2. local maximum: $D > 0$ and $\partial^2 f / \partial x^2 < 0$
3. saddle point: $D < 0$
4. undetermined: $D = 0$

Constrained Optimization

(제약 최적화)



with a constraint (제약 조건)

Example 8.15

Find the extremum value of the function $f = e^{-x^2-y^2}$ subject to the constraint $x + y = 1$.

The constraint imposes $y = 1 - x$ and, hence,

$$f = e^{-x^2-y^2} = e^{-x^2-(1-x)^2} = e^{-2x^2+2x-1}$$

Now this is a function of one variable, so at a minimum/maximum,

$$\frac{df}{dx} = (-4x + 2)e^{-2x^2+2x-1} = 0$$

The only solution is $x = 1/2$, which yields $y = 1 - x = 1/2$.

Hence, the extremum value is $f(1/2, 1/2) = e^{-1/2}$.

Lagrange's Method

Find the constrained extremum in $f(x, y)$,
when the constraint is written in the form $g(x, y) = 0$

1. Form the new function

$$u(x, y) = f(x, y) + \lambda g(x, y)$$

where λ is a constant called an **undetermined multiplier**
(미정승수).

Lagrange's Method

Find the constrained extremum in $f(x, y)$,
when the constraint is written in the form $g(x, y) = 0$

1. New function $u(x, y) = f(x, y) + \lambda g(x, y)$
2. Set the partial derivatives of u equal to zero:

$$\left(\frac{\partial u}{\partial x}\right)_y = \left(\frac{\partial f}{\partial x}\right)_y + \lambda \left(\frac{\partial g}{\partial x}\right)_y = 0$$

$$\left(\frac{\partial u}{\partial y}\right)_x = \left(\frac{\partial f}{\partial y}\right)_x + \lambda \left(\frac{\partial g}{\partial y}\right)_x = 0$$

Lagrange's Method

Find the constrained extremum in $f(x, y)$,
when the constraint is written in the form $g(x, y) = 0$

1. New function $u(x, y) = f(x, y) + \lambda g(x, y)$
2. Partial derivatives of $u = 0$
3. Solve the equations from point 2 and $g(x, y) = 0$
to obtain x , y , and $\lambda \rightarrow$ constrained extrema

Example 8.16

Find the constrained extremum of the function $f = e^{-x^2-y^2}$ subject to the constraint $x + y = 1$, using Lagrange's method.

The constraint equation is $g(x, y) = x + y - 1 = 0$

The new function u is $u(x, y) = f(x, y) + \lambda g(x, y) = e^{-x^2-y^2} + \lambda(x + y - 1)$

The three equations to be solved are

$$\left(\frac{\partial u}{\partial x}\right)_y = (-2x)e^{-x^2-y^2} + \lambda = 0$$

$$\left(\frac{\partial u}{\partial y}\right)_x = (-2y)e^{-x^2-y^2} + \lambda = 0$$

$$g(x, y) = x + y - 1 = 0$$

Example 8.16

Find the constrained extremum of the function $f = e^{-x^2-y^2}$ subject to the constraint $x + y = 1$, using Lagrange's method.

$$(-2x)e^{-x^2-y^2} + \lambda = 0$$

$$(-2y)e^{-x^2-y^2} + \lambda = 0$$

$$x + y - 1 = 0$$

From the first two equations, $x = y$

Substitution into the third equation gives $x + x - 1 = 0$, or $x = 1/2$

Hence, at an extremum, $x = y = 1/2$ and its value is $e^{-1/2}$.

Function with Three Variables

For a function $f = f(x, y, z)$ subject to the two constraints $g_1(x, y, z) = 0$ and $g_2(x, y, z) = 0$, use two undetermined multipliers:

$$u(x, y, z) = f(x, y, z) + \lambda_1 g_1(x, y, z) + \lambda_2 g_2(x, y, z)$$

A similar procedure leads to the constrained extrema

Example 8.17

Find the extremum of the function $f = x^2 + y^2 + z^2$ subject to the constraints $y = 1$ and $z = 2$, using Lagrange's method.

The constraint equations are

$$g_1(x, y, z) = y - 1 = 0 \text{ and } g_2(x, y, z) = z - 2 = 0$$

The new function u is

$$\begin{aligned} u(x, y, z) &= f(x, y, z) + \lambda_1 g_1(x, y, z) + \lambda_2 g_2(x, y, z) \\ &= x^2 + y^2 + z^2 + \lambda_1(y - 1) + \lambda_2(z - 2) \end{aligned}$$

Its partial derivatives lead to

$$(\partial u / \partial x)_{y,z} = 2x = 0$$

$$(\partial u / \partial y)_{x,z} = 2y + \lambda_1 = 0$$

$$(\partial u / \partial z)_{x,y} = 2z + \lambda_2 = 0$$

Example 8.17

Find the extremum of the function $f = x^2 + y^2 + z^2$ subject to the constraints $y = 1$ and $z = 2$, using Lagrange's method.

$$\begin{aligned} 2x &= 0, & 2y + \lambda_1 &= 0, & 2z + \lambda_2 &= 0 \\ & & y &= 1, & z &= 2 \end{aligned}$$

We have five variables and five equations,
but it is straightforward to solve them:

$$x = 0, \quad y = 1, \quad z = 2, \quad \lambda_1 = -2, \quad \lambda_2 = -4$$

Hence, the extremum value is

$$f(0, 1, 2) = 0^2 + 1^2 + 2^2 = 5.$$

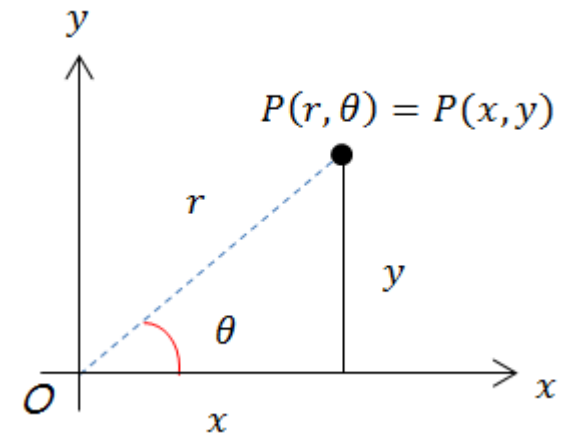
Coordinate Systems

- 2 dimensions

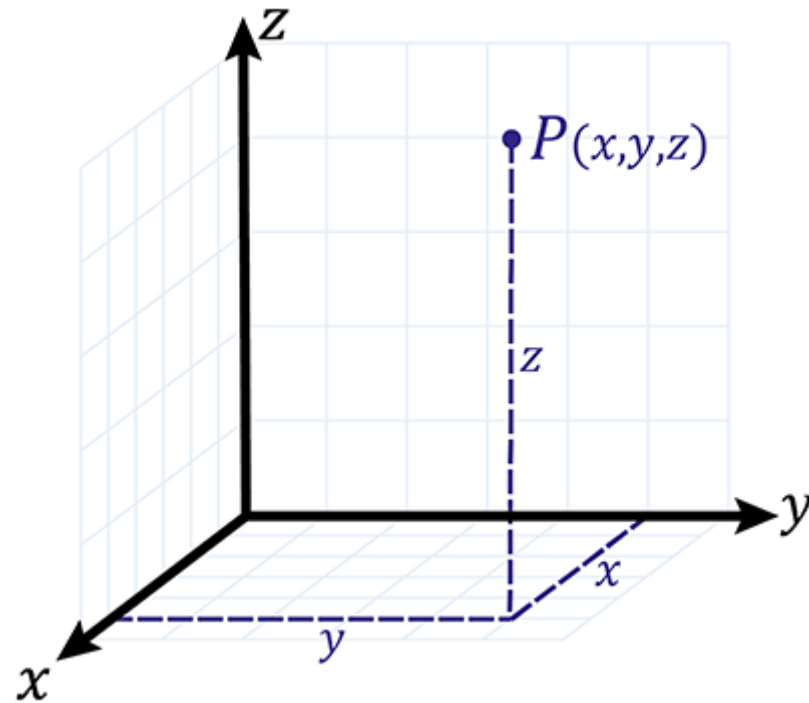
- Cartesian coordinates (직교 좌표)
- Polar coordinates (극좌표)

- 3 dimensions

- Cartesian coordinates (직교 좌표)
- Spherical (polar) coordinates (구면 좌표)
- Cylindrical (polar) coordinates (원통 좌표)

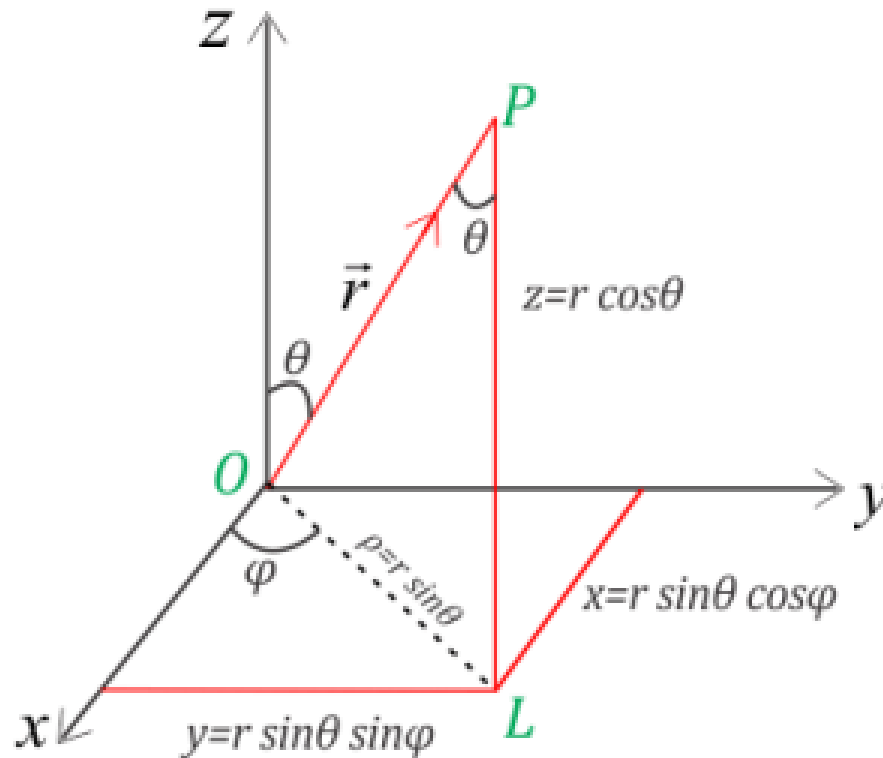


Cartesian Coordinates in 3 Dimensions



(right-handed coordinate system)

Spherical Polar Coordinates



$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1}(z/r)$$

$$\phi = \tan^{-1}(y/x)$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

Example 3.3

Find the spherical polar coordinates of the point whose Cartesian coordinates are $(1, 1, 1)$.

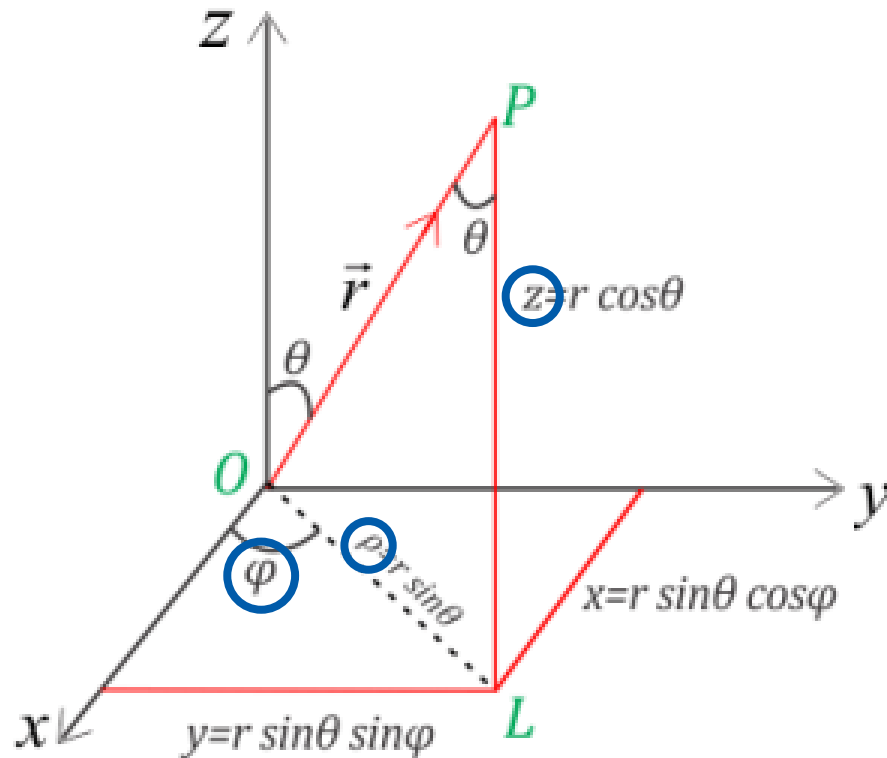
Using the previous formulas,

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\theta = \cos^{-1}(z/r) = \cos^{-1}(1/\sqrt{3}) \approx 54.7^\circ$$

$$\phi = \tan^{-1}(y/x) = \tan^{-1}(1/1) = \tan^{-1} 1 = 45^\circ$$

Cylindrical Polar Coordinates



$$\rho = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}(y/x)$$

$$z = z$$

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$

(Image: <https://physicscatalyst.com/graduation/spherical-coordinates-system/>)

Example 3.4

Find the cylindrical polar coordinates of the point whose Cartesian coordinates are $(1, -4, -2)$.

Using the previous formulas,

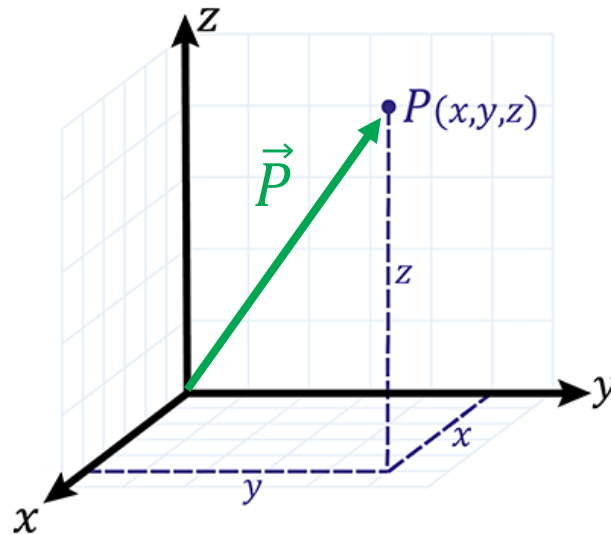
$$\rho = \sqrt{x^2 + y^2} = \sqrt{1^2 + 4^2} = \sqrt{17}$$

$$\phi = \tan^{-1}(y/x) = \tan^{-1}(-4/1) = \tan^{-1}(-4) \approx 284^\circ$$

$$z = -2$$

Vector

A quantity with magnitude and direction



(Image: <https://www.skillsyouneed.com/num/cartesian-coordinates.html>)

Representation of Vector

For a vector $\vec{P}$ in 3 dimensions,

- Vector: $\vec{P}$ or $\mathbf{P}$
- Vector components: P_x, P_y, P_z

$$\vec{P} \leftrightarrow (P_x, P_y, P_z)$$

- In a different coordinate system, we can use

$$\vec{P} \leftrightarrow (P_r, P_\theta, P_\phi)$$

Different Representations

Consider a vector $\vec{P}$ whose

- Cartesian components are (1,1,1)

$$P_x = 1, P_y = 1, P_z = 1$$

- Spherical components are $(\sqrt{3}, 54.7^\circ, 45^\circ)$

$$P_r = \sqrt{3}, P_\theta = 54.7^\circ, P_\phi = 45^\circ$$

Unit Vectors

$\hat{i} = \mathbf{i}$: vector of unit length pointing x^+

$\hat{j} = \mathbf{j}$: vector of unit length pointing y^+

$\hat{k} = \mathbf{k}$: vector of unit length pointing z^+

In the Cartesian coordinate system,

$$\hat{i} \leftrightarrow (1,0,0), \quad \hat{j} \leftrightarrow (0,1,0), \quad \hat{k} \leftrightarrow (0,0,1)$$

Hence, any vector $\vec{P}$ can be represented as

$$\vec{P} = \hat{i}P_x + \hat{j}P_y + \hat{k}P_z$$

Magnitude of Vector

For a vector $\vec{A} = \hat{i}A_x + \hat{j}A_y + \hat{k}A_z$, its **magnitude** is

$$|\vec{A}| = A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Example 4.5

Find the magnitude of the vector $\vec{A} = (3,4,5)$.

Using the definition,

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50}$$

Vector Operations

For $\vec{A} = \hat{i}A_x + \hat{j}A_y + \hat{k}A_z$ and $\vec{B} = \hat{i}B_x + \hat{j}B_y + \hat{k}B_z$,

- Sum

$$\vec{A} + \vec{B} = \hat{i}(A_x + B_x) + \hat{j}(A_y + B_y) + \hat{k}(A_z + B_z)$$

- Difference

$$\vec{A} - \vec{B} = \hat{i}(A_x - B_x) + \hat{j}(A_y - B_y) + \hat{k}(A_z - B_z)$$

- Multiplication by a scalar c

$$c\vec{A} = \hat{i}(cA_x) + \hat{j}(cA_y) + \hat{k}(cA_z)$$

Scalar Product (내적)

The scalar product (dot product) of $\vec{A}$ and $\vec{B}$ is

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

and, equivalently,

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \alpha$$

where α is the angle between the two vectors.

Example 4.6

Let $\vec{A} = 2\hat{i} + 3\hat{j} + 7\hat{k}$ and $\vec{B} = 7\hat{i} + 2\hat{j} + 3\hat{k}$. Find $\vec{A} \cdot \vec{B}$ and the angle between $\vec{A}$ and $\vec{B}$.

From the first definition,

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z = 2 \cdot 7 + 3 \cdot 2 + 7 \cdot 3 = 41$$

To obtain the angle, we can use $\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos \alpha$; the magnitudes are

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{2^2 + 3^2 + 7^2} = \sqrt{62}$$

$$|\vec{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{7^2 + 2^2 + 3^2} = \sqrt{62}$$

Hence, $\cos \alpha = \vec{A} \cdot \vec{B} / |\vec{A}||\vec{B}| = 41/62$ and $\alpha = \cos^{-1}(41/62) = 48.6^\circ$

Vector Product (외적)

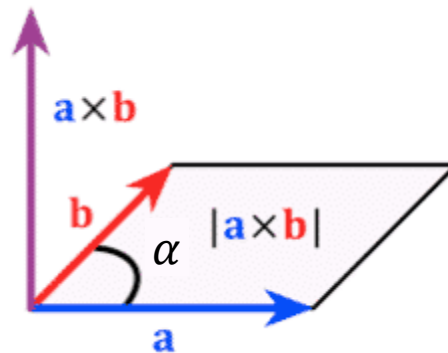
The vector product (cross product) of $\vec{A}$ and $\vec{B}$ is

$$\begin{aligned}\vec{A} \times \vec{B} &= \hat{i}(A_y B_z - A_z B_y) + \hat{j}(A_z B_x - A_x B_z) + \hat{k}(A_x B_y - A_y B_x) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}\end{aligned}$$

Geometric Definition of Vector Product

For $\vec{C} = \vec{A} \times \vec{B}$,

- Magnitude: $|\vec{C}| = |\vec{A}||\vec{B}| \sin \alpha$ (α : angle between $\vec{A}$ & $\vec{B}$)
- Direction: right-hand rule



Example 4.7

Find the vector product $\vec{A} \times \vec{B}$, where $\vec{A} = (1,2,3)$ and $\vec{B} = (1,1,1)$.

From the first definition,

$$\vec{A} \times \vec{B} = \hat{i}(A_y B_z - A_z B_y) + \hat{j}(A_z B_x - A_x B_z) + \hat{k}(A_x B_y - A_y B_x)$$

$$\vec{A} \times \vec{B} = \hat{i}(2 \cdot 1 - 3 \cdot 1) + \hat{j}(3 \cdot 1 - 1 \cdot 1) + \hat{k}(1 \cdot 1 - 2 \cdot 1) = -\hat{i} + 2\hat{j} - \hat{k}$$